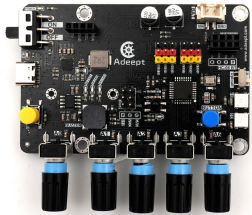


Lesson 1 Light up the LED

1.1 Overview

This course mainly introduces how to use Adeept Robotic Arm Control Board to control onboard LED lights. Through simple code writing and hardware connection, the LED lights can flash at fixed time intervals and their status can be viewed in real-time on the serial monitor.

1.2 Required Components

Components	Quantity	Picture
Adeept Robotic Arm Control Board	1	
Type-C USB Cable	1	

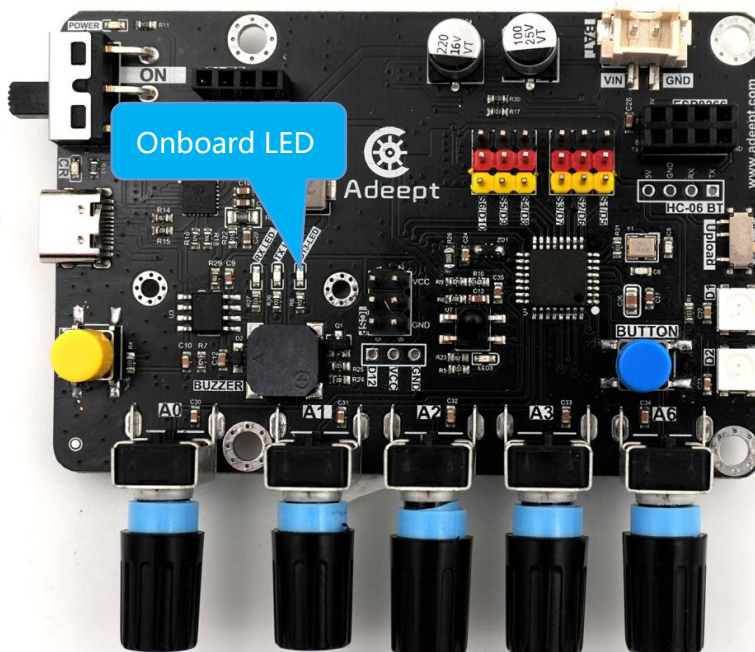
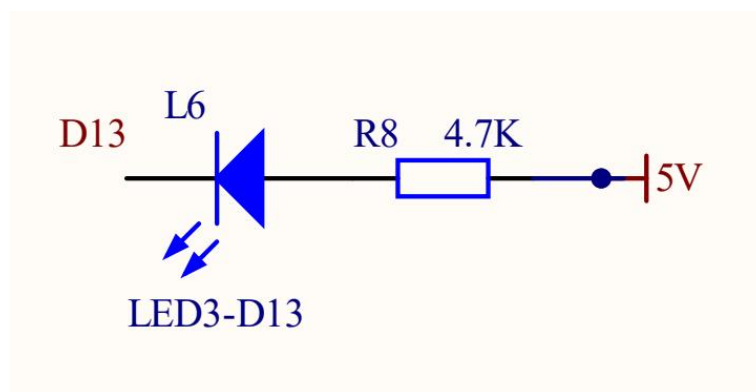
1.3 Principle Introduction

The D13 pin on the Adeept Robotic Arm Control Board is connected to the onboard LED. When outputting a high level (HIGH) to pin D13, the LED lights up; When the output is low (LOW), the LED light goes out. By controlling the voltage state of the D13 pin in the code, it is possible to control the on/off of the LED light.

PINS of Adeept Robotic Arm Control Board	LED
D13	LED

1.4 Wiring Diagram

The onboard LED is connected to the D13 pin.



1.5 Demonstration

1. Connect your computer and Adeeps Robotic Arm Control Board with a USB cable.

2. Open "01_LED" folder in "/Code" , double-click "01_LED.ino" .

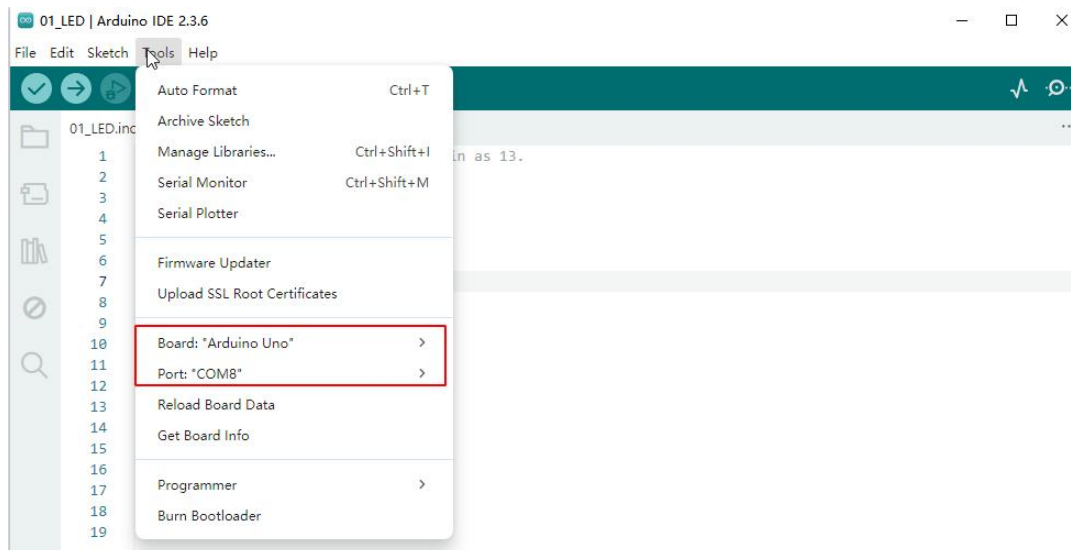


3. Select development board and serial port.

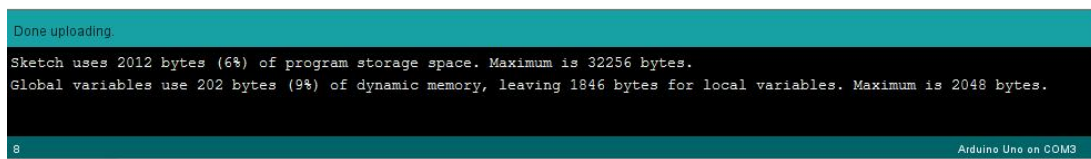
Board: **Tools**--->**Board**--->**Arduino AVR Boards**--->**Arduino Uno**

Port: **Tools** --->**Port**--->**COMx**

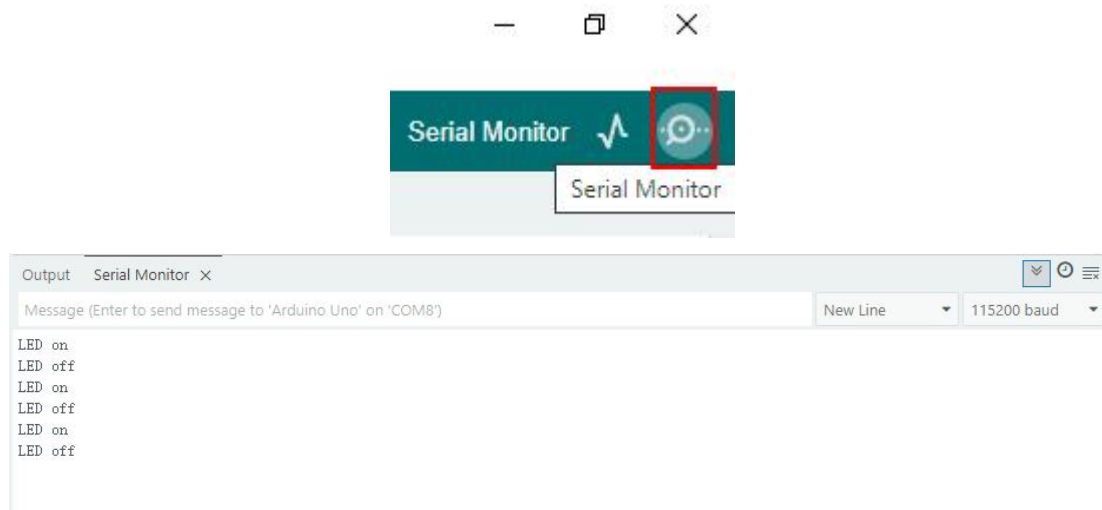
Note: The port number will be different in different computers.



4. After opening, click  to upload the code program to the Arduino. If there is no error warning in the console below, it means that the Upload is successful.



5. Click Serial Monitor, Set the baud rate as 115200.



6. After successfully running the program, the onboard LED will blink once every 1s.

1.6 Code

Complete code refer to [01_LED.ino](#)

```
01 # define PIN_LED 13 // Define LED pin as 13.
02
03 void setup()
04 {
05     Serial.begin(115200);
06     pinMode(PIN_LED, OUTPUT);
07     delay(10);
08 }
09
10 void loop()
11 {
12     digitalWrite(PIN_LED, LOW);
13     Serial.println("LED on");
14     delay(1000);
15     digitalWrite(PIN_LED, HIGH);
16     Serial.println("LED off");
17     delay(1000);
18 }
```